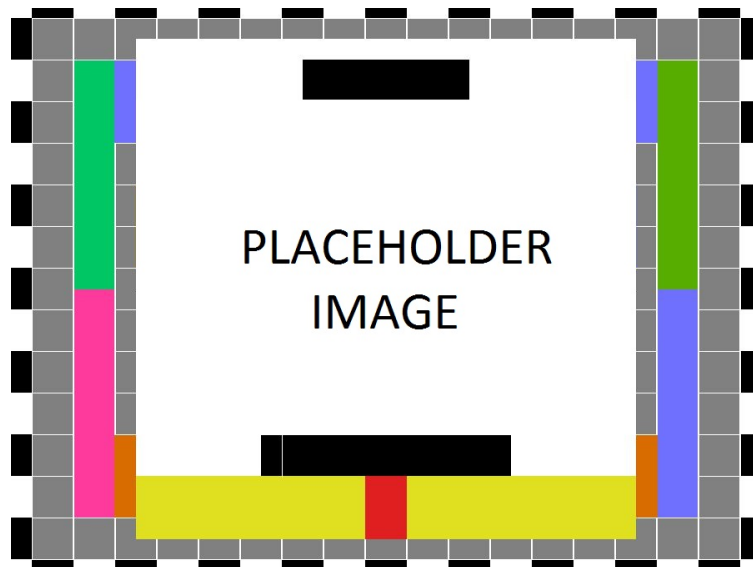


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**SYSTEM REQUIREMENTS SPECIFICATION
CSE 4316: SENIOR DESIGN I
FALL 2026**



**TEAM FULLSTACK
CIVICPULSE COMMUNITY SERVICE-REQUEST
DASHBOARD**

**ALAN TURING
GRACE HOPPER
JOHN VON NEUMANN**

ADA LOVELACE
CHARLES BABBAGE

REVISION HISTORY

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1 INTRODUCTION

This SRS specifies CivicPulse Community Service-Request Dashboard, a web application for community service requests. It follows ISO/IEC/IEEE 29148:2018 [1].

1.1 PURPOSE

Agreed requirements baseline between Team Fullstack and the city.

1.2 SCOPE

The system shall let residents submit and track geolocated service requests and let authorized staff triage, assign, update, and close them, with resident status notifications. Field-crew mobile apps and payment processing are out of scope.

1.3 PRODUCT OVERVIEW

1.3.1 PRODUCT PERSPECTIVE

A three-tier web application (browser frontend, API backend, database) integrating with the city SSO (OIDC), a geocoding/mapping API, and an SMTP relay.

1.3.2 PRODUCT FUNCTIONS

Request submission with location/photo; resident status tracking; staff triage, assignment, and status update; duplicate detection; email notifications; reporting.

1.3.3 USER CHARACTERISTICS

Residents (unauthenticated or lightly registered), staff agents, and staff administrators (role-based).

1.3.4 ASSUMPTIONS & DEPENDENCIES

City SSO test realm; provided categories/routing; free-tier geocoding; SMTP relay.

1.4 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

Term	Definition
Request	A resident-reported service issue with type, location, and status
Triage	Staff review that validates, categorizes, and prioritizes a request
OIDC	OpenID Connect single sign-on
RBAC	Role-Based Access Control

2 REQUIREMENTS ENGINEERING APPROACH

Identifiers [TYPE] - [NN] (EI, FR, US, PE, SE, MS, QA) with the 29148 attribute set. Requirements are singular and verifiable.

3 EXTERNAL INTERFACE REQUIREMENTS

3.1 EI-01: CITY SSO (OIDC)

DESCRIPTION

Staff authentication shall be delegated to the city identity provider via OpenID Connect authorization-code flow.

RATIONALE

Staff must use existing city credentials; the app must not store staff passwords.

SOURCE

Sponsor IT.

PRIORITY

Critical

VERIFICATION METHOD

Test — staff sign-in via the SSO realm.

CONSTRAINTS

Depends on the SSO test realm.

APPLICABLE STANDARDS

OpenID Connect Core 1.0, OAuth 2.0 (RFC 6749).

3.2 EI-02: GEOCODING/MAPPING API

DESCRIPTION

The system shall geocode addresses and render request locations on a map via an external mapping API.

RATIONALE

Locations must be normalized and visualized for triage.

SOURCE

Team.

PRIORITY

High

VERIFICATION METHOD

Demonstration — submit an address; see it mapped.

CONSTRAINTS

Free-tier rate limits.

APPLICABLE STANDARDS

None.

4 FUNCTIONAL REQUIREMENTS

4.1 FR-01: SUBMIT REQUEST

DESCRIPTION

A resident shall submit a request with category, description, location (map pin or address), and optional photo, and receive a tracking id.

RATIONALE

Intake is the entry point of the whole workflow.

SOURCE

Sponsor.

PRIORITY

Critical

VERIFICATION METHOD

Test — submit and receive a tracking id.

CONSTRAINTS

Photo size limited; location required.

APPLICABLE STANDARDS

None.

4.2 FR-02: TRACK REQUEST

DESCRIPTION

A resident shall view the current status and history of a request by its tracking id without authentication.

RATIONALE

Self-service status is the main resident benefit.

SOURCE

Sponsor.

PRIORITY

Critical

VERIFICATION METHOD

Demonstration — look up a tracking id.

CONSTRAINTS

No PII of other residents exposed.

APPLICABLE STANDARDS

None.

4.3 FR-03: STAFF TRIAGE & ASSIGNMENT

DESCRIPTION

An authenticated agent shall filter/sort the request queue, set priority, and assign a request to a crew or department.

RATIONALE

Staff need to organize and route the backlog.

SOURCE

Sponsor.

PRIORITY

Critical

VERIFICATION METHOD

Demonstration — triage and assign a request.

CONSTRAINTS

Restricted to the agent role (SE-01).

APPLICABLE STANDARDS

None.

4.4 FR-04: STATUS UPDATE & NOTIFICATION

DESCRIPTION

When an agent changes a request status, the system shall record the change and email the resident (if contact provided).

RATIONALE

Closes the loop and reduces status-check calls.

SOURCE

Sponsor.

PRIORITY

High

VERIFICATION METHOD

Test — change status; verify record + email.

CONSTRAINTS

Requires SMTP relay.

APPLICABLE STANDARDS

None.

4.5 FR-05: DUPLICATE DETECTION

DESCRIPTION

On submission the system shall flag likely duplicates (same category within a configurable radius/time) for staff review.

RATIONALE

Cuts duplicate handling effort.

SOURCE

Sponsor.

PRIORITY

Moderate

VERIFICATION METHOD

Test — near-duplicate submissions are flagged.

CONSTRAINTS

Radius/time configurable.

APPLICABLE STANDARDS

None.

5 USABILITY REQUIREMENTS

5.1 US-01: ACCESSIBLE, RESPONSIVE UI

DESCRIPTION

The resident portal and staff dashboard shall conform to WCAG 2.2 AA and be usable on mobile and desktop viewport widths (360–1920 px).

RATIONALE

Public services must be accessible to all residents on any device.

SOURCE

Sponsor; law.

PRIORITY

High

VERIFICATION METHOD

Test — automated + manual accessibility audit.

CONSTRAINTS

Modern browsers.

APPLICABLE STANDARDS

WCAG 2.2 AA.

6 PERFORMANCE REQUIREMENTS

6.1 PE-01: RESPONSIVENESS

DESCRIPTION

Interactive pages shall respond within 500 ms (p95) and support at least 200 concurrent users on the staging host.

RATIONALE

Sluggish public services are abandoned.

SOURCE

Sponsor.

PRIORITY

High

VERIFICATION METHOD

Test — load test at target concurrency.

CONSTRAINTS

Bounded by host.

APPLICABLE STANDARDS

None.

7 SECURITY & PRIVACY REQUIREMENTS

7.1 SE-01: RBAC & TRANSPORT SECURITY

DESCRIPTION

All staff functions shall be gated by role, and all traffic shall use TLS 1.3; plaintext HTTP shall redirect to HTTPS.

RATIONALE

Prevents unauthorized access and eavesdropping on resident data.

SOURCE

Sponsor; security review.

PRIORITY

Critical

VERIFICATION METHOD

Test — role checks; TLS scan.

CONSTRAINTS

On staging and production.

APPLICABLE STANDARDS

OWASP ASVS; RFC 8446 (TLS 1.3).

7.2 SE-02: RESIDENT DATA MINIMIZATION

DESCRIPTION

The system shall collect only the resident contact data needed for notifications and shall not expose personal data on public tracking pages.

RATIONALE

Privacy obligation under the city policy.

SOURCE

Sponsor policy.

PRIORITY

High

VERIFICATION METHOD

Inspection — data-flow and page review.

CONSTRAINTS

City privacy policy.

APPLICABLE STANDARDS

None.

8 MAINTENANCE & SUPPORT REQUIREMENTS

8.1 MS-01: OBSERVABILITY

DESCRIPTION

The system shall emit structured logs and health metrics sufficient to diagnose failures and monitor uptime without code changes.

RATIONALE

Operable by city IT after handover.

SOURCE

Team.

PRIORITY

Moderate

VERIFICATION METHOD

Inspection.

CONSTRAINTS

Log retention sized for staging.

APPLICABLE STANDARDS

ISO/IEC 25010:2023 (analysability) [2].

9 OTHER QUALITY ATTRIBUTES

9.1 QA-01: AVAILABILITY

DESCRIPTION

The service shall achieve $\geq 99.5\%$ monthly availability at the public endpoint.

RATIONALE

Residents rely on it being reachable.

SOURCE

Sponsor.

PRIORITY

High

VERIFICATION METHOD

Analysis — uptime measurement.

CONSTRAINTS

Single-region staging.

APPLICABLE STANDARDS

ISO/IEC 25010:2023 (availability) [2].

10 VERIFICATION & TRACEABILITY

Req ID	Requirement (abbrev.)	Method	Acceptance
FR-01	Submit request	T	tracking id re-returned
FR-04	Status + notification	T	record + email sent
US-01	Accessibility	T	WCAG 2.2 AA pass
PE-01	Responsiveness	T	p95 ≤500 ms @200 users
SE-01	RBAC + TLS	T	role checks; TLS1.3

11 FUTURE ITEMS

11.1 FI-01: NATIVE FIELD-CREW MOBILE APP

DESCRIPTION

An offline-capable mobile app for field crews to update requests on-site.

RATIONALE

Valuable but beyond prototype scope.

SOURCE

Sponsor.

PRIORITY

Future

VERIFICATION METHOD

Test (if implemented).

CONSTRAINTS

Requires mobile dev + offline sync.

APPLICABLE STANDARDS

None.

REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 29148:2018 — Systems and software engineering — Life cycle processes — Requirements engineering,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2018.
- [2] ISO/IEC, “ISO/IEC 25010:2023 — Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2023.